sCure / CureBox — Design Verification Test (DVT)

Campaign overview · catalog v0.23 · SRS-DVT-SW Rev B · 2026-09-10

40	5	517	≈850 h	6	69
TESTS	UNITS UNDER TEST	APPLICABLE RUNS	BENCH TIME	RUNS COMMITTED	CONFIRM ITEMS OPEN

1 · WHAT THE DVT PROVES

The DVT qualifies the sCure resin-curing chamber design on five production-representative units before release: **electrical safety** (bonding, insulation, hipot, touch current, inrush), **every protective function** under injected faults (door interlock, thermal cutout, watchdog, sensor and fan failures, mains faults, LED and nitrogen faults), **performance** (chamber thermal characterisation, LED temperature, cool-down, nitrogen purge), **transport** and, last, two **destructive** mains-fault tests on the designated unit. Tests by subsystem: Safety 21 · Process 8 · Thermal 5 · Electrical 5 · Environmental 1.

Every test is defined once, in a machine-readable catalog (YAML) that the DVT console executes: applicability per unit, sweep or case matrix, equipment with calibration status, preconditions, step-by-step procedure with the fields each step records, data fields with plausible ranges, and a pass criterion evaluated automatically (PASS / FAIL / BLOCKED — a criterion whose limit is still undecided is reported BLOCKED, never PASS).

2 · PHASES AND GATES

PHASE	TESTS	RUNS	HOURS	GATE TO THE NEXT PHASE
0 — Incoming and configuration	0	0	0	Configuration frozen and signed; equipment calibrations valid
1 — Electrical safety	5	40	48	All ELE tests PASS
2 — Protective functions	25	319	559	All protective functions PASS or approved disposition
3 — Performance	6	155	238	Data complete
4 — Transport	1	0	0	Post-shipment functional test PASS
5 — Destructive	3	3	6	—

A phase opens for a unit only when the previous one is closed for that unit; protective earth (ELE-001) must PASS before anything is energised. Phase 0 is the configuration freeze (serial, firmware, catalog revision) and the readiness review (TRR) signed per unit.

3 · UNITS

UNIT	ROLE	NOTES	RUNS DONE
UUT-01	reference	SAF-010 review, SAF-018 software stack	3/114
UUT-02	standard		2/100
UUT-03	standard		1/100
UUT-04	standard		0/100
UUT-05	destructive	THM-004 / SAF-001 / SAF-002 — last	0/103

4 · METHOD — THE RULES EVERY TEST FOLLOWS

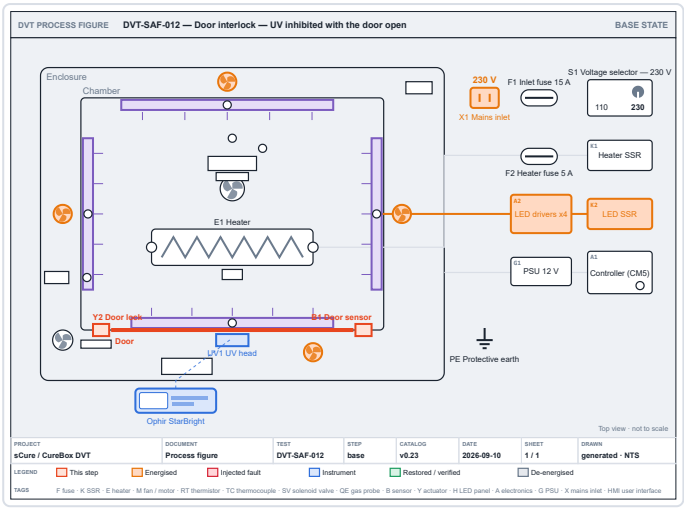
- **Shutdowns are proven on the output, not the controller.** A heater cut-off is read from the type-K thermocouple bonded to the heater body (PicoLog TC-08): the instant the trace turns from rising to falling is when power was removed; faults are injected during heat-up at full output so the turn is unmistakable. LED cut-off is read from the Ophir StarBright irradiance trace (below 10 % of plateau). The controller's own report is recorded, never trusted alone.
- **Alarms must name the component.** Every fault test records the alarm text and code and judges whether it names the right fan, sensor, panel or line — and, where the machine cannot know (a blocked air path vs a weak fan), whether it says so instead of guessing.
- **Every unit for workmanship, one unit for design.** Bonding, seals, harnesses and fittings vary unit to unit and are tested on all five; software behaviour and destructive faults run once on the designated unit.
- **Nothing is guessed.** Limits still to be decided are `CONFIRM` items in the catalog; the affected tests stay BLOCKED until the number is set and the decision is logged.

5 · INSTRUMENTATION

- **PicoLog TC-08 ×2** — 16 type-K thermocouples on a fixed 3-D grid (THM-001), heater-body and LED MCPCB bonds; the TC-08 clock is the time base of every shutdown test.
- **Ophir StarBright** UV power meter (PD300-UV head) read directly by the console: LEDs at maximum → trigger → irradiance vs time, graph and CSV attached to the run (SAF-012 door, SAF-007 thermistor, SAF-014 watchdog).
- **Oxygen analyser** (centre + worst corner), inline N_2 flow meter, chamber manometer — purge to 95 % N_2, hold through the cure, inlet over-pressure, venting, room O_2 at the operator.
- **Programmable AC source** 100 / 110 / 230 / 240 V, true-RMS meter, clamp on the heater branch, current probe with single-shot scope for the fuse tests.
- **Safety bench** — earth-bond tester, insulation tester, hipot, touch-current meter; thermal camera and fire-rated enclosure for the destructive tests.

6 · THE DVT CONSOLE

- **Guided execution** — one wizard per run: safety plan, equipment and calibration check, preconditions, each step with only its fields, live values from the machine, redlines, attachments, verdict; a restart resumes where it stopped.
- **Machine in the loop** — the console talks to the unit's hardware service (state, heat / cool / UV / door / diagnostics) and its SSH host; a built-in simulator with fault injection rehearses every protective test without hardware.
- **Team work through Microsoft SharePoint** — the campaign folder is a SharePoint document library (Microsoft 365) synced to each bench with OneDrive; each console keeps its own journal there and replays the others' every 15 s: one campaign, several benches, first claim on a run wins; attachments in the shared folder.
- **Controlled history in Git** — a checkpoint (campaign JSON, workbook, report, summary) is committed at every phase gate and on demand.
- **Outputs** — dashboard with progress per subsystem and per machine, verdict matrix across units, temperature statistics per component, XLSX / CSV / Markdown exports, the full campaign document (every test with its engineering process figure), NCRs and waivers with approver and rationale.



Process figure — one generated engineering sheet per test and per step: the machine, the mains chain, the instruments, the injected fault, with component tags and a title block (here: DVT-SAF-012, door interlock).

7 · STATUS

Catalog v0.23 baselined for review: 40 tests, 517 applicable runs, ≈850 bench hours. Runs committed so far: **6 / 517** · 1 PASS · 1 FAIL · 4 BLOCKED · 1 open NCRs.

8 · DECISIONS NEEDED BEFORE PHASE 2

Limits still null in the catalog: `led_temperature.warning`, `led_temperature.protective_shutdown`, `heater.selector_mismatch_fuse_open_max`, `nitrogen.purge_time_max_s`, `nitrogen.hold_time_min_s`, `nitrogen.max_inlet_pressure_bar`, `nitrogen.chamber_pressure_limit_mbar`, `nitrogen.coil_temp_limit_c`. The tests that reference them evaluate as BLOCKED until set.

- **Fan and LED feedback.** No tach on the second fan of each pair and no per-panel LED output feedback: SAF-008 / 019 / 020 need a hardware or firmware decision (tach lead, per-panel current sense on the spare ADS1115 inputs, or a thermistor rise-rate rule).
- **Nitrogen sensing.** No line-pressure sensor on the board (`n2LinePressure = null`); the 6 bar enable guard and the pre-cure N_2 check cannot work, and the Settings guard blocks nitrogen mode on real hardware. Fit an inlet pressure switch or drop the guard.
- **Voltage selector.** A mid position is an open circuit; only the 25-minute stall rule can catch it and its text does not mention the selector. A "no rise at all" rule with customer guidance is recommended.
- **REQ-48 heat-up ≤ 20 min** already fails at EVT (27–33 min); decide the number or a documented waiver before Phase 3. Low line at 100 V is now in THM-001; 207 V still open.
- **Campaign logistics** — the destructive unit (UUT-05), the shipped unit for ENV-001, witnessing of Phases 1 and 5, a climate-controlled room for THM-003.